

EC3218 — Portfolio Analysis

Summer 2022 Examination · Paper 1 · Full Marking Scheme

Module	EC3218 — Portfolio Analysis
Internal Examiner	Dr. Jun Gao
Institution	University College Cork
Session	Summer 2022, Paper 1
Duration	1.5 hours
Structure	Section A — Long Questions: attempt 2 of 3 (100% of the paper). All questions carry equal weight.
Materials	Non-programmable calculator permitted

Coverage note. This scheme answers **all three** Section A questions in full, since you'll choose your best two under exam conditions.

Own-figure rule. A1(c) and A2(c) are multi-step numerical questions — an answer that applies the correct method consistently from an earlier arithmetic slip still earns the marks available for the later steps.

Source basis. The behavioural-finance error taxonomy in A2(a) follows the module's own material (information-processing errors vs. behavioural/preference biases). The specific classification of each named scenario in A2(a)(iii)-(iv) follows the standard behavioural finance literature (Kahneman and Tversky's representativeness heuristic and prospect theory) since the exact scenarios weren't set out verbatim in the uploaded course pack — flagged at that question.

Section A · Question 1

Bonds and stocks, trading volume as evidence, and two-asset frontiers — 100 marks

A1(a)

[25 marks]

Describe the basic characteristics and types of bonds and stocks.

A1(b)

[20 marks]

"One piece of evidence of inefficient markets can be found in the volume of trading activity. The volume of trading always increases before a takeover announcement." Evaluate this statement.

A1(c)

[55 marks]

You are given the following information: Stock A, Expected Return $E(R)=8$, Risk $\sigma=6$; Stock B, Expected Return $E(R)=5$, Risk $\sigma=4$. (i) Plot the efficient and inefficient frontiers if the correlation between A and B is 1. (ii) Plot the efficient and inefficient frontiers if the correlation between A and B is -1 . (iii) Plot the efficient and inefficient frontiers if the correlation between A and B is -0.5 .

FULL MODEL ANSWER

Bonds — basic characteristics: a bond is a **debt security** — the issuer borrows a fixed principal (the **par/face value**) from the bondholder and promises to repay it at a stated future date (**maturity**), typically along with periodic interest payments (the **coupon**). Bondholders are **creditors**, not owners: they have a prior legal claim on the issuer's cash flows and assets ahead of shareholders, and no ongoing say in how the business is run, in exchange for a contractually fixed (rather than residual) return.

- **Government bonds:** issued by national governments (e.g. Irish or German government bonds), generally considered very low default risk for creditworthy sovereigns.
- **Corporate bonds:** issued by companies, carrying genuine default (credit) risk that varies with the issuer's financial strength — reflected in credit ratings and, correspondingly, the yield demanded by investors.
- **Zero-coupon bonds:** pay no periodic coupon at all, sold at a discount to par and redeemed at full face value at maturity, with the investor's entire return coming from that price appreciation.

Stocks — basic characteristics: a stock (equity share) represents **fractional ownership** in a company — shareholders are residual claimants, entitled to whatever remains after all other obligations (including bondholders) are met, with no guaranteed payment at all. This makes equity a **higher-risk, higher-expected-return** claim than debt in the same company.

- **Common stock:** the standard form of equity ownership, typically carrying voting rights and a residual claim on dividends and assets, with no fixed or guaranteed return.
- **Preferred stock:** a hybrid instrument — typically pays a fixed dividend (like a bond's coupon) and usually carries no voting rights, but sits behind bondholders and ahead of common shareholders in the priority of claims, making it intermediate in risk between debt and common equity.

Mark Allocation — Part (a) (25 marks)

Tier	Marks	What separates this tier
19-25	19-25	Correctly describes bonds as debt claims (with maturity, coupon, priority) and stocks as residual equity claims AND names and correctly distinguishes at least two types of each.
13-18	13-18	Correct basic characteristics given for both, with at least one type of each named.
7-12	7-12	Only one of bonds/stocks correctly and fully described.
0-6	0-6	No credible description given.

Why increased pre-announcement trading volume is often cited as evidence of inefficiency: if trading volume in a target company's stock reliably rises *before* a takeover is publicly announced, this is consistent with **information leakage** — some market participants trading on advance, non-public knowledge of the impending deal. Since a takeover announcement is exactly the kind of major, price-moving **private** information that strong-form efficiency says should not be tradeable on ahead of public release, systematic pre-announcement volume spikes are a natural candidate piece of evidence against strong-form efficiency specifically.

Why the statement needs careful evaluation rather than acceptance at face value: the claim as stated — that volume "always" increases beforehand — is too strong and conflates two genuinely different explanations that would have very different implications for market efficiency.

- **Genuine information leakage/insider trading:** if the volume increase is driven by trading on material non-public information (leaked deal terms, advance knowledge from insiders or their associates), this is a real violation of strong-form efficiency — exactly the kind of pattern documented by Jaffe (1974) and others for insider trading more generally.
- **Rational anticipation based on public information:** takeover targets are often not random — they may already show public signs consistent with being an attractive target (depressed valuations relative to fundamentals, activist investor stakes, public rumour and speculation, industry consolidation trends). A rise in volume driven by *public* speculation and analysis, rather than private knowledge of a specific deal, is entirely consistent with semi-strong (and even weak-form) efficiency — it reflects the market rationally processing publicly available signals, not a violation of anything.

The evaluation: the statement is **not fully correct as an unqualified claim**. A pre-announcement volume increase is *consistent with*, but does not on its own *prove*, market inefficiency — establishing genuine information leakage (rather than rational public anticipation) requires additional evidence, such as unusual trading concentrated among parties with plausible access to the deal (advisors, executives, their associates), which is exactly the kind of evidence real insider-trading investigations and enforcement actions rely on, rather than the volume pattern alone.

Mark Allocation — Part (b) (20 marks)

Tier	Marks	What separates this tier
15-20	15-20	Correctly identifies the information-leakage interpretation as evidence against strong-form efficiency AND correctly identifies the alternative, efficiency-consistent explanation (rational anticipation from public signals) AND gives a balanced evaluation noting the statement's "always" is too strong without further evidence distinguishing the two.
10-14	10-14	Correctly identifies the information-leakage interpretation, but the alternative explanation or the balanced evaluation is missing.
5-9	5-9	Simply agrees the statement is evidence of inefficiency without any critical evaluation.
0-4	0-4	No credible evaluation given.

Setting up the two-asset opportunity set: for two assets with weights w_a and $w_B = 1 - w_a$, portfolio risk and return are given by $\sigma_p = \sqrt{w_a^2 \sigma_a^2 + w_B^2 \sigma_B^2 + 2w_a w_B \rho \sigma_a \sigma_B}$ and $E(R_p) = w_a E(R_a) + w_B E(R_B)$. The shape of the resulting frontier as w_a varies from 0 to 1 depends entirely on the correlation, ρ .

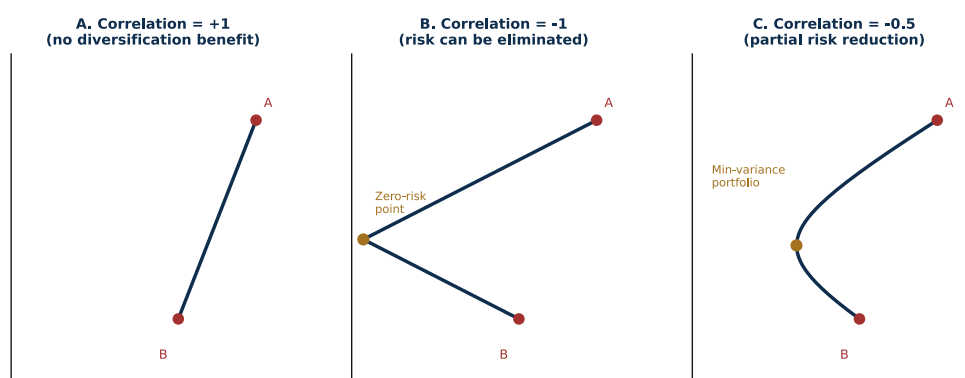


Fig. 1 — The two-asset opportunity set at $\rho = +1$ (a straight line, no diversification benefit), $\rho = -1$ (risk can be driven all the way to zero), and $\rho = -0.5$ (a genuine curved frontier, offering partial risk reduction).

(i) Correlation = +1 — no diversification benefit at all: with perfect positive correlation, portfolio risk collapses to the simple weighted average $\sigma_p = w_a \sigma_a + w_B \sigma_B$ — a **straight line** connecting B ($\sigma = 4$, $E(R) = 5$) to A ($\sigma = 6$, $E(R) = 8$). Since risk and return move in perfect lockstep along this line (every increase in expected return requires an equal proportional increase in risk, with no offsetting reduction possible), the **entire line is the efficient frontier** — there is no inefficient segment, since diversification provides no risk-reduction benefit whatsoever when $\rho = 1$.

(ii) Correlation = -1 — risk can be driven all the way to zero: with perfect negative correlation, it becomes possible to combine A and B in exactly the right proportion to eliminate portfolio risk entirely. Solving $w_a \sigma_a = w_B \sigma_B$ gives $w_a = \sigma_B / (\sigma_a + \sigma_B) = 4/10 = 0.4$ ($w_B = 0.6$), at which $\sigma_p = 0$ and $E(R_p) = 0.4(8) + 0.6(5) = 6.2$. The frontier forms two straight rays meeting at this zero-risk point: the **efficient** ray runs from ($\sigma = 0$, $E(R) = 6.2$) up to A, and the **inefficient** ray runs from the same zero-risk point down to B — for any given risk level, the ray toward A always offers a strictly higher return than the ray toward B.

(iii) Correlation = -0.5 — genuine, partial risk reduction: with

correlation strictly between -1 and $+1$, the frontier becomes a genuine curve (a hyperbola), offering real but incomplete diversification benefit. The minimum-variance weight is $w_a^* = (\sigma_B^2 - \rho\sigma_a\sigma_B)/(\sigma_a^2 + \sigma_B^2 - 2\rho\sigma_a\sigma_B) = \mathbf{0.368}$ ($w_B = 0.632$), giving a minimum-variance portfolio of $\sigma_p = \mathbf{2.38}$, $E(R_p) = \mathbf{6.11}$ — a risk level below *either* individual asset (below B's own $\sigma=4$), but not driven all the way to zero as it was with $\rho=-1$. The **efficient frontier** runs from this minimum-variance point up to A; the **inefficient** portion runs from the same minimum-variance point down to B.

EXAM TECHNIQUE

The general pattern across all three parts: the **lower** the correlation, the **greater** the diversification benefit (the more the curve bows toward the origin, reducing risk below what either asset offers alone) — $\rho=+1$ is the worst case (no benefit at all) and $\rho=-1$ is the best case (risk can be eliminated completely); $\rho=-0.5$ sits between the two extremes, exactly as the diagram shows.

Mark Allocation — Part (c) (55 marks)

Tier	Marks	What separates this tier
42-55	42-55	All three cases correctly plotted/described AND the zero-risk point for $\rho=-1$ ($w_a=0.4$, $E(R)=6.2$) and the minimum-variance point for $\rho=-0.5$ ($w_a=0.368$, $\sigma=2.38$, $E(R)=6.11$) both correctly calculated AND the efficient vs. inefficient segments correctly identified in each case AND the general pattern (lower correlation $\rightarrow$ greater diversification benefit) is stated.
28-41	28-41	All three shapes qualitatively correct, but at least one of the two specific numerical points (zero-risk point; minimum-variance point) is incorrectly calculated.
14-27	14-27	Only the $\rho=+1$ and $\rho=-1$ cases correctly handled; the $\rho=-0.5$ case (which requires the full minimum-variance-weight formula) is missing or incorrect.
0-13	0-13	No credible frontier calculations attempted.

Question A1 Total: 100 Marks

Section A · Question 2

Behavioural finance errors, market risk and diversification, and a long/short pairs trade — 100 marks

A2(a)

[40 marks]

(i) Which two types of errors are specific to the Behavioural Finance Theory? (ii) Which of the above are not really errors, and why? (iii) If an investor reads on the Internet that a particular firm has sold more products this year and, as a

result, he buys stock in this firm, which type of error in judgment does he make? (iv) An investor did not earn the return he wanted this year; in fact, he even lost money. He receives information about a firm that may soon be quoted on a stock exchange. Based on this information he decides to buy stock in this firm in the hope he will improve his poor financial position. Which type of error of preference does he make?

A2(b)

[30 marks]

"Diversification enables us to reduce some of the risk in a portfolio, but market risk will always remain." Do you agree with this statement? Justify your answer using appropriate economic reasons.

A2(c)

[30 marks]

Suppose you are interested in two stocks, A and B. The spot prices of A and B are €15 per share. You anticipate the price of A to increase to €20, but the price of B to decline to €10 in one year. You have €9,000 and plan to borrow 100 shares of B from a stockbroker. Explain your investment strategy and calculate the expected return of your investment strategy.

FULL MODEL ANSWER

(i) The two types of errors specific to Behavioural Finance Theory: the standard framework divides behavioural departures from rationality into two distinct categories:

- **Information processing errors:** mistakes investors make in *estimating* probabilities and expected returns from available information — forecasting errors (over-extrapolating from limited recent evidence), overconfidence, conservatism (underreacting to new evidence), and sample size neglect/representativeness (treating a small sample as fully representative of a larger pattern).
- **Behavioural (preference) biases:** systematic departures in how investors *evaluate* and choose between outcomes, even when the underlying probabilities are correctly assessed — framing (decisions affected by how a choice is presented), mental accounting (treating money differently depending on which mental "account" it's assigned to), regret avoidance, and affect (letting how an investment makes you *feel* influence the decision, independent of risk and return).

(ii) Which are not really errors, and why: the **behavioural/preference biases** (framing, mental accounting, regret avoidance, affect) are the more debatable category — arguably not "errors" at all in the strict sense, because they can reflect a coherent, stable set of personal preferences rather than an objective mistake in judgement. An investor who mentally separates a "safe" retirement account from a "speculative" account (mental accounting), or who derives genuine utility from investing in line with personal values (affect), isn't necessarily miscalculating anything — they may simply have a richer utility function than pure risk-return optimisation captures. **Information processing errors**, by contrast, are much harder to defend as anything other than genuine mistakes — overconfidence and conservatism involve objectively miscalculating probabilities or failing to update on new evidence correctly, which cannot be reframed as a legitimate alternative preference.

(iii) Buying stock after reading online that the firm sold more products this year: this is best classified as a **representativeness / sample size**

neglect error (an information-processing error) — the investor is treating a single, recent data point (one year's higher sales) as fully representative of the firm's ongoing prospects, extrapolating a pattern from limited evidence without properly weighing whether this is a genuine, durable trend or just one-off noise. This is closely related to the module's own "forecasting errors" category — giving too much weight to recent experience relative to the broader, more uncertain picture.

(iv) The investor who lost money and buys a speculative stock hoping to recover losses: this is a **framing error** (a behavioural/preference bias), specifically rooted in **Prospect Theory** (Kahneman and Tversky, 1979): investors are typically **risk-averse when facing gains** but **risk-seeking when facing losses**, evaluated relative to a personal reference point (here, the return the investor originally wanted or expected). Having already lost money relative to that reference point, the investor is psychologically in the "loss domain," which — per prospect theory — makes taking on a risky, speculative bet in an attempt to "get back to even" *more* attractive than it would otherwise be, purely because of how the decision is framed relative to the investor's reference point, not because of any genuine reassessment of the new stock's risk-adjusted prospects.

Mark Allocation — Part (a) (40 marks)

Tier	Marks	What separates this tier
31-40	31-40	Correctly names both error categories (information processing; behavioural/preference biases) AND correctly identifies preference biases as the "not really errors" category with reasoning AND correctly classifies scenario (iii) as representativeness/sample size neglect AND correctly classifies scenario (iv) as a framing error rooted in prospect theory's risk-seeking-in-losses result.
21-30	21-30	Both categories correctly named and at least one of the two scenarios correctly classified with reasoning.
11-20	11-20	Categories named, but neither scenario correctly classified, or the "not really errors" question not addressed.
0-10	0-10	No credible answer given.

Agree — diversification eliminates only unsystematic risk; systematic (market) risk is, by definition, common to all assets and cannot be diversified away.

The economic reasoning: total risk in any individual asset can be split into two components: **unsystematic (firm-specific/idiosyncratic) risk** — factors affecting one company alone (a factory fire, a product recall, a lawsuit, management turnover) — and **systematic (market) risk** — factors affecting essentially every asset simultaneously, arising from common macroeconomic forces (interest rate changes, recessions, inflation shocks, broad shifts in investor risk appetite).

Why diversification eliminates only the unsystematic component: unsystematic shocks are, by their nature, largely **independent** across different companies — one firm's factory fire has no direct bearing on another, unrelated firm's earnings. Combining many such independent risks in a portfolio causes them to substantially offset each other on average (see the EC3005 Winter 2025 scheme, Q. A1(a), on how the same logic underlies risk pooling in insurance), which is exactly why portfolio variance falls as more distinct assets are added. **Systematic risk cannot be diversified away by the same logic**, precisely because it is *not* independent across assets — a

recession or an interest-rate shock moves virtually every asset in the same direction at the same time, so adding more assets to the portfolio does nothing to offset it; the common factor affects every holding simultaneously, regardless of how many different assets are held.

The practical implication: this is precisely why CAPM's beta — a measure of an asset's exposure to *systematic* risk specifically — is the risk measure that determines expected return in equilibrium, rather than an asset's total standard deviation: rational, diversified investors are only compensated for bearing the risk they *cannot* eliminate through diversification (systematic risk), since unsystematic risk can be eliminated for free simply by holding a sufficiently diversified portfolio, and the market has no reason to reward investors for bearing a risk they didn't need to take on in the first place.

Mark Allocation — Part (b) (30 marks)

Tier	Marks	What separates this tier
23-30	23-30	Correctly agrees AND correctly distinguishes systematic from unsystematic risk AND correctly explains why independence across firms lets diversification eliminate unsystematic risk but not the shared, common-factor-driven systematic risk AND connects this to why beta (not total risk) determines expected return in equilibrium.
15-22	15-22	Correctly agrees with the systematic/unsystematic distinction given, but the independence-based reasoning for why diversification only works on the unsystematic component is not explained.
8-14	8-14	States that "some risk always remains" without the systematic/unsystematic distinction or reasoning.
0-7	0-7	Disagrees without valid justification, or no credible answer given.

The investment strategy — a levered long/short pairs trade: since A is expected to **rise** and B is expected to **fall**, the strategy is to **go long A** (buy, expecting to profit from the price rise) and **short B** (borrow and sell now, expecting to profit by buying it back cheaper later) — a classic pairs trade that profits from the expected *divergence* between the two stocks, using the short sale to help fund the long position rather than tying up additional capital.

Step	Calculation	Amount
Short 100 shares of B at spot €15	$100 \times €15$	+€1,500 (proceeds received)
Total funds available to invest in A	€9,000 (own) + €1,500 (short proceeds)	€10,500
Shares of A purchased	€10,500 ÷ €15	700 shares

Explaining the mechanics: the €1,500 raised by short-selling the borrowed B shares is added to the investor's own €9,000, giving €10,500 in total buying power — all of which is used to buy 700 shares of A at €15. No additional personal capital beyond the original €9,000 is required, since the short sale itself supplies the extra funding — this is exactly what makes the strategy **levered**: the investor ends up controlling €10,500 of long exposure to A using only €9,000 of their own capital.

Calculating the outcome in one year: close both positions at the anticipated prices — sell the A shares at their new price, and buy back the B shares at their new (lower) price to return them to the broker.

Step	Calculation	Amount
Value of 700 A shares at €20	$700 \times €20$	€14,000
Cost to buy back 100 B shares at €10 (closing the short)	$100 \times €10$	–€1,000
Net cash at year-end	€14,000 – €1,000	€13,000

The expected return: profit = €13,000 (final cash) – €9,000 (original own capital) = **€4,000**. Since only the investor's own €9,000 was actually committed as capital (the B shares were borrowed, not purchased), the **expected return on own capital** is $€4,000 \div €9,000 = 44.44\%$ — substantially higher than either stock's own price change in isolation (A alone would have returned 33.3%), precisely because the strategy is leveraged: both the long gain on A *and* the short gain on B contribute to profit, on a capital base smaller than the total position size.

Mark Allocation — Part (c) (30 marks)

Tier	Marks	What separates this tier
23-30	23-30	Correctly identifies the strategy as a levered long-A/short-B pairs trade AND correctly calculates the 700 shares of A purchased using combined own capital and short proceeds AND correctly calculates the year-end profit of €4,000 AND correctly calculates the 44.44% return on own capital specifically (not on the total position size).
15-22	15-22	Strategy correctly identified and share count correctly calculated, but an error in the final profit or return calculation.
8-14	8-14	Strategy correctly identified in words, but the numerical calculation is substantially incorrect.
0-7	0-7	No coherent strategy or calculation given.

Question A2 Total: 100 Marks

Section A · Question 3

Efficient markets and analysis, and the efficient frontier with and without a risk-free asset
— 100 marks

A3(a)

[40 marks]

(i) What is an efficient market? Describe some important implications of the efficient market theory for technical and fundamental analysis. (ii) Briefly discuss the three forms of the efficient market theory.

A3(b)**[60 marks]**

With the aid of a diagram, answer the following questions: (i) What is meant by the term Efficient Frontier? (Assume no short sales and no riskless lending and borrowing is allowed.) (ii) Explain how the efficient frontier is expected to change if riskless borrowing and lending is allowed. (iii) 'The mean-variance efficiency criterion can identify the optimal investment'. Do you agree or disagree? Justify your answer using appropriate economic reasons.

FULL MODEL ANSWER

(i) What an efficient market is: a market in which asset prices **fully and rapidly reflect all available information** — Fama (1970): prices "fully reflect" available information, built on the empirical finding that stock prices follow a **random walk** (Kendall, 1953), with price changes appearing statistically independent from one period to the next.

Implications for technical analysis: technical analysis relies entirely on identifying patterns in **past** prices and trading volume to predict future price movements. If even the **weak form** of the EMH holds, all such past information is already fully incorporated into current prices — meaning any genuinely exploitable pattern would already have been traded away by the market. The implication is stark: **technical analysis cannot generate abnormal returns** if weak-form efficiency holds, since by the time a pattern is identifiable, it's already reflected in the price.

Implications for fundamental analysis: fundamental analysis uses **publicly available** information (financial statements, industry data, macroeconomic conditions) to estimate a security's intrinsic value. If the **semi-strong form** holds, all such public information is already incorporated into prices the moment it becomes available, meaning fundamental analysis based purely on public information **cannot systematically generate abnormal returns** either. However, fundamental analysis retains a more defensible role than technical analysis even under semi-strong efficiency: it is arguably the very **mechanism** by which public information gets impounded into prices in the first place (see the EC3218 Summer 2023 scheme, Q. A1(c), on the Grossman-Stiglitz paradox) — an analyst with genuinely superior interpretive skill, or access to private information, could still earn abnormal returns even under semi-strong (though not strong-form) efficiency.

(ii) The three forms

Form	Claim
Weak	Prices reflect all past price and volume information — technical analysis cannot generate abnormal returns
Semi-strong	Prices reflect all past AND public information — fundamental analysis of public information alone cannot generate abnormal returns
Strong	Prices reflect all information, public AND private — not even insider information can generate abnormal returns (the evidence mostly rejects this form)

Mark Allocation — Part (a) (40 marks)

Tier	Marks	What separates this tier
31-40	31-40	Correct definition of an efficient market with the random walk foundation

AND correctly explains the implication for technical analysis (no abnormal returns possible if weak form holds) **AND** correctly explains the implication for fundamental analysis (no abnormal returns from public information alone under semi-strong form, but a role remains in producing that efficiency) **AND** all three forms correctly and briefly described.

21-30	21-30	Correct definition and both analysis-type implications given, but only briefly, or the three forms are named without their specific claims.
11-20	11-20	Definition given with only one of the two analysis-type implications correctly explained.
0-10	0-10	No credible answer given.

(i) The Efficient Frontier, without short sales or riskless

lending/borrowing: the **Minimum Variance Frontier** plots, for each achievable level of expected portfolio return, the smallest possible risk obtainable from some combination of the available risky assets (with no negative weights allowed, since short sales are ruled out here). The **efficient frontier** is the **upper portion** of this curve, from the minimum-variance portfolio upward — for every point on the lower portion, a portfolio directly above it on the upper portion offers the same risk with a strictly higher expected return, making the lower portion dominated and therefore never rational to hold.

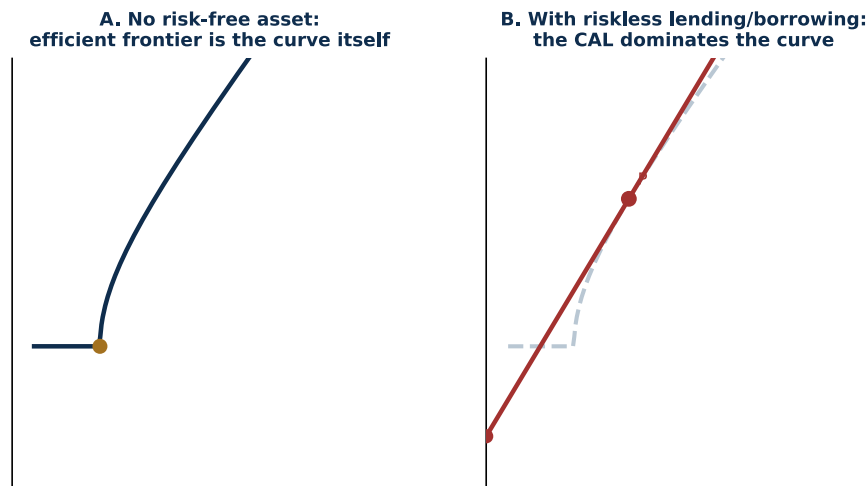


Fig. 1 — Panel A: without a risk-free asset, the efficient frontier is the curve itself. Panel B: once riskless lending and borrowing are allowed, the Capital Allocation Line — tangent to the curve at portfolio P — dominates the curved frontier everywhere except at the single point of tangency.

(ii) How the frontier changes once riskless lending and borrowing is

allowed: the opportunity set expands dramatically. A **Capital Allocation Line (CAL)** can now be drawn from the risk-free rate on the vertical axis, tangent to the curved efficient frontier at exactly one point — the **tangency portfolio, P**. This straight CAL **dominates every point on the original curved frontier** except at the single tangency point itself, since combining the risk-free asset with P in varying proportions achieves a strictly better risk-return trade-off (a higher Sharpe ratio, the CAL's slope) than any risky-asset-only combination on the curve could. The **new efficient frontier becomes the CAL itself**: risk-averse investors hold more of the risk-free asset and less of P (lending); less risk-averse investors can even borrow at the risk-free rate to hold *more* than 100% of their wealth in P — extending the achievable frontier out beyond P itself along the same straight line.

Disagree, or at best only partially agree — mean-variance efficiency narrows the choice down to the efficient set, but does not by itself identify a single optimal portfolio for any given investor.

Why mean-variance efficiency alone cannot identify THE optimal investment: the mean-variance criterion is genuinely useful for eliminating *dominated* portfolios — anything not on the efficient frontier (or CAL, once a risk-free asset is available) can be ruled out for any rational investor. But this leaves an entire **frontier** (or, with a risk-free asset, an entire line) of candidate portfolios remaining, not a single point — the actual choice of *where* along that frontier or line an individual investor should sit depends entirely on that investor's own **risk preferences (utility function/indifference curves)**, which the mean-variance criterion, by itself, says nothing about. Two investors facing the exact same efficient frontier can rationally choose very different portfolios along it, and mean-variance efficiency alone cannot say which one is "the" optimal choice.

A further, deeper caveat: the mean-variance framework itself rests on restrictive assumptions — either that returns are normally distributed (so that mean and variance fully describe the return distribution) or that investor utility is quadratic (so that only the first two moments matter to the investor at all). If an investor genuinely cares about skewness or tail risk beyond what variance captures — plausible for many real investors, especially around large, asymmetric downside risks — mean-variance efficiency can fail to identify what is truly optimal for that investor even after accounting for individual risk preference, since the criterion itself may be omitting risk dimensions the investor actually cares about.

Mark Allocation — Part (b) (60 marks)

Tier	Marks	What separates this tier
46-60	46-60	Efficient frontier without a risk-free asset correctly defined and diagrammed AND the change with riskless lending/borrowing (CAL dominates the curve, tangency portfolio) correctly explained and diagrammed AND a well-reasoned disagreement with part (iii) given, correctly identifying that individual risk preferences (not just the efficient set) are needed to pin down a single optimal portfolio, ideally with the further caveat about the framework's own distributional/utility assumptions.
31-45	31-45	Both diagram parts (i) and (ii) correctly given, but part (iii)'s reasoning is only partially developed (e.g. states disagreement without explaining why risk preferences are the missing ingredient).
16-30	16-30	Only part (i) or (ii) correctly given, with a weak or absent answer to part (iii).
0-15	0-15	No credible diagram or discussion given.

Question A3 Total: 100 Marks